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WENO interpolation for Lagrangian particles in highly compressible flow regimes

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ABSTRACT

Lagrangian particles can be used either to analyze different complex flows, as massless tracers, or to model multiphase flows, as particles with mass. As particles can have arbitrary positions within the computational domain, interpolation of different flow quantities from the Eulerian grid is essential. Low-order centered interpolation schemes generally do not provide a sufficient level of accuracy for many flow configurations of interest. Thus, typically, high-order centered interpolation schemes are utilized. The current study demonstrates that for highly compressible non-reacting and reacting flow regimes, in which discontinuities in the flow field, e.g., shock waves arise, centered interpolation schemes tend to smear the shock and high order schemes also produce numerical oscillations. It is shown that this problem can be remedied by using Weighted-Essentially-Non-Oscillatory (WENO) interpolation schemes. Extensive numerical tests are performed in order to demonstrate this, comparing the performance of WENO-3 and WENO-5 interpolation schemes with a variety of centered interpolation schemes. The first test case involves specially designed steady state two-dimensional vortex flows. For a smooth flow, WENO schemes provide comparable accuracy to other high-order centered schemes. For flow fields with discontinuities, WENO-3 and WENO-5 interpolation schemes can decrease the interpolation error by more than two and three orders of magnitude, respectively, in comparison with centered schemes. Solution accuracy is further studied with a normal shock wave test. It is demonstrated that centered schemes tend to smear the shock, whereas, WENO schemes capture it in a much sharper manner. Moreover, high-order centered schemes tend to oscillate in the vicinity of the discontinuity leading to non-physical results, such as negative absolute pressure values. The same trends are retained for an unsteady three-dimensional spherical blast wave, where the shock is smeared across several grid cells, which is typical for shock-capturing flow solvers. Finally, the benefits of WENO schemes for the Lagrangian-particle tracking analysis of highly compressible reactive flows are explored by comparing various Lagrangian particle trajectories and joint probability density functions (PDFs) for a two-dimensional cellular detonation. The results indicate that high-order centered schemes lead to unphysical results, whereas WENO schemes can provide high-order interpolation that is free of severe numerical oscillations and non-physical artifacts, which is critical for the proper analysis of the flow.

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1. Introduction

A wide range of flows involve high speeds and compressibility. These include supersonic and hypersonic boundary layers [1], atmospheric flow around the reentry vehicles [2], as well as various propulsion and energy conversion systems, such as scramjet engines, where the combustion process is supersonic [3], and pulsating and rotating detonation engines [4,5]. Other examples include industrial explosions such as in coal mines [6] or storage silos [7]. High-speed regimes can also arise through the complex interaction between exothermic reactions and turbulence that entails significant up-scale transfer of kinetic energy known as backscatter [8–10]. For instance, interaction between flames and high-speed turbulence can lead to a deflagration-to-detonation transition (DDT) [11].

Numerical simulations of these complex flows often involve Lagrangian particles that can be either massless or with a finite mass. Massless particle trajectories can be used to investigate the flow field, in a procedure known as Lagrangian-particle tracking analysis. On the other hand, particles with mass can be used to model multiphase flows, for instance, droplets in spray combustion [12]. Numerical solution of the coupled set of flow and Lagrangian particle equations requires at least two key steps. First, as particle position rarely coincides with the Eulerian flow field cell centers, interpolation is necessary in order to reconstruct different flow quantities at a particle position. Second, a numerical time integration scheme is utilized in order to advect the particles across the grid. For particles with mass, a third step is necessary, as two-way coupling between the particles and the fluid requires feedback of momentum and energy from particles position to the cell centers. For highly compressible flow regimes that involve discontinuities in the flow, such as shock waves, special treatment may be necessary for each of these three steps, in order to attain an accurate and physically sound numerical solution. The current work focuses on the first step in order to tackle this complicated problem. Thus, here we consider solely the operation of interpolation for grid quantities.

As mentioned above, one possible method to investigate complex flows is via the Lagrangian-particle tracking analysis, where a large number of point massless particles (fluid parcels) are seeded in the flow and subsequently their trajectories are followed. The only difference of this representation of the flow field from the classical Eulerian point of view, which is typically used to model flows, is the frame of reference. Lagrangian analysis, however, has several advantages, especially for complex multi-physics flows, e.g., with reactions [13]. First, different quantities of interest such as pressure, temperature, mass fraction of species, etc., can be tracked directly through the flow. This is contrary to the Eulerian frame of reference, where these quantities are defined only at discrete points. Second, as a Lagrangian particle exactly follows the flow trajectory, there is no advection into or out of the particle. This allows one to separate the effect of complex multi-physics processes, e.g., chemical reaction or molecular transport, from the convective transport. Third, particle residence times and path lengths during different stages of interest of the flow evolution can be calculated. All of these attributes of the Lagrangian analysis can provide significant insights into the flow dynamics, not achievable in the Eulerian frame of reference. One should note that proper analysis is only possible once the temporal and spatial numerical errors associated with time integration and interpolation, respectively, are sufficiently small.

Lagrangian analysis of flows based on the direct numerical simulations (DNS) was pioneered more than 40 years ago by Riley and Patterson [14]. They analyzed a non-reacting incompressible turbulent flow, where a low-order linear interpolation was utilized to reconstruct the velocity at the Lagrangian particle positions. Yeung and Pope [15] conducted a comprehensive analysis in order to determine the required level of accuracy for both time integration and spatial interpolation methods. They found that time integration errors are negligible as typically the time-step size used for DNS simulations is rather small. This stems from the fact that for many flow solvers the Courant–Friedrichs–Lewy (CFL) number must be lower than 1 to achieve a stable solution. Spatial interpolation errors, on the other hand, were found to be more significant, and the use of higher-order interpolation methods was recommended, as low-order linear interpolation was found not sufficiently accurate. Since then, many studies have tested the accuracy of different interpolation schemes for non-reacting incompressible turbulent flows [16–18], and today it is agreed that low-order linear interpolation cannot provide a sufficient level of accuracy [19]. For a detailed discussion of the Lagrangian analysis of non-reacting incompressible turbulent flows we refer the readers to the extensive review papers by Yeung [19] or Toschi and Bodenschatz [20].

For more complex, multi-physics flows, in particular with chemical reactions, Lagrangian-particle tracking analysis has started to emerge only in recent years. Day et al. [21] used Lagrangian analysis to investigate premixed turbulent methane flames in a low-swirl combustor. Linear interpolation was used in order to reconstruct different Eulerian field quantities at a particle position [22]. In a follow-up paper, the same configuration was studied with premixed turbulent hydrogen flames [23]. Darragh et al. [24] developed a Lagrangian-tracking particle method for reacting flows. They tested the accuracy of the method using different time integration and interpolation schemes. It was also concluded that linear interpolation does not provide sufficient accuracy, and the chosen methods were high-order Akima splines for spatial interpolation in conjunction with the 4th- and 5th-order explicit Runge–Kutta methods for time integration. Hamlington et al. [13] utilized this framework to study high-speed premixed turbulent hydrogen-air flame with single- and multi-step reaction kinetics. Fukushima et al. [25] used DNS with Lagrangian analysis in order to classify combustion regimes for the homogeneous charge compression ignition (HCCI) and premixed charge compression ignition (PCCI) engines. They interpolated values at a particle position via the 4th-order Lagrange polynomial interpolation and utilized the 3rd order Runge–Kutta method for time integration. Another application that was recently pursued involved the use of the Lagrangian analysis in order to gain a better understanding of premixed autoignition [26]. It is also worth mentioning recent works that consider Lagrangian tracking of flame elements [27,28] and Lagrangian analysis using experimental data sets [29–31]. One should note that reacting flows typically have variable density, as they involve large density changes due to the expansion of gas in flame. At the same time, for all studies mentioned above, the compressibility levels of the flow are rather low. For instance, Day et al. [21] are using the low-Mach approximation in order to solve the flow equations. Yet, even when fully compressible flow solvers are utilized, as in Ref. [13], the flows themselves are relatively weakly compressible with a turbulent Mach number not exceeding the value of 0.3 in any of the studied cases. Moreover, none of these works deal with highly shocked flows or detonations, and thus it is not surprising that the same methods used for non-reacting incompressible turbulent flows are applicable.

For the Lagrangian analysis of non-reacting compressible turbulent flows, the approaches used are different. Most studies use linear interpolation [32–39], and some of them describe difficulties that arise once high-order interpolation schemes are utilized. For instance, Beetz et al. [32] report difficulties with the interpolation of density, which must always be positive. Yet, they describe negative density values near strong shocks [33]. Yang et al. [34] note that a linear interpolation is used in order to avoid unphysical oscillations near shocklets, which suggests that higher-order interpolation methods tend to oscillate. It should be noted that Federrath et al. [36] and Konstandin et al. [37] report that higher-order schemes do not improve the accuracy of their results, but do not describe any specific issues with high-order interpolation. Obviously, as for the case of incompressible turbulence, high-order interpolation can provide higher level accuracy at least for some applications, and allows one to avoid the need for excessive grid refinement to reduce the spatial interpolation error. Works that deal with both reacting and highly compressible flows are scarce. For instance, in Ref. [40], nonpremixed autoignition is investigated for turbulent Mach numbers as high as unity. Yet, no special treatment for interpolation from the Eulerian grid to the Lagrangian particle position is performed.

The issue of numerical oscillations, which are induced by high-order schemes, discussed above is actually well-known in other quite different fields. For instance, in smooth particle hydrodynamics (SPH), kernel interpolation is performed typically with 2nd-order schemes, which has the same order of accuracy as linear interpolation. The reason is that high-order schemes tend to oscillate and undershoot leading to negative density values [41]. A more common example is the problem faced by compressible flow solvers once shocked flows arise. It is known that low-order methods will never oscillate in the presence of shocks [42]. However, low-order methods are extremely diffusive, inaccurate, and tend to smear the shock, which requires excessive grid refinement that is not always practical. High-order methods are much more accurate, but without any special treatment they will always oscillate near the shock in a rather unphysical manner [42]. Different solutions to this problem have been suggested over the years, e.g., total variation diminishing (TVD) methods [43].

To the best of the authors' knowledge, a solution for this issue for massless Lagrangian point particles has not been demonstrated. For particles with mass, however, Jacobs and Don [44] suggested to use an Essentially-Non-Oscillatory (ENO) interpolation scheme. The idea behind ENO is that the interpolation stencil is divided into several sub-stencils and one stencil is chosen according to its local smoothness [45]. They showed that ENO interpolation can provide a more accurate and stable solution than high-order centered interpolation schemes. In the current work, we suggest to solve the issue of numerical oscillations induced by high-order centered interpolation schemes by utilizing a Weighted-Essentially-Non-Oscillatory (WENO) interpolation scheme. WENO schemes were first developed by Liu et al. [46] and then improved by Jiang and Shu [47]. WENO interpolation is similar to ENO interpolation, and the interpolation procedure is based on a convex combination of all candidate sub-stencils instead of a single sub-stencil like in the case of ENO. This approach has several advantages. First, the approach adopted by Jacobs and Don [44] requires switching between a centered interpolation scheme, once the flow is smooth, and an ENO scheme, once the flow is discontinuous. This means that some threshold criterion for switching has to be chosen, and its specific value can affect the solution. For WENO schemes, no threshold condition is necessary, and the scheme automatically reduces to a centered scheme once the flow is smooth. Second, it is difficult to vectorize ENO schemes as they are based on multiple conditional statements. WENO schemes are based mostly on arithmetic operations, and thus can be vectorized in an efficient manner. Third, the ENO implementation presented by Jacobs and Don [44] cannot handle multiple shocks that cross the stencil. The proposed WENO formulation can handle multiple shocks with some limitations, as will be described below. One should note that both ENO and WENO schemes do not eliminate oscillations completely. However, the magnitude of the oscillations decays as $O(\Delta x^k)$, where Δx is the grid spacing and k is the order of accuracy [45]. Thus, oscillations are greatly reduced as the grid is refined, and for any practical purpose become negligible. This is actually the reason these schemes are referred to as essentially non-oscillatory schemes.

Kozak et al. [48] presented a WENO-3 interpolation scheme for Lagrangian point particles and tested its performance against different centered interpolation schemes. The main focus of that work was on the coupling of different interpolation schemes with symplectic or trajectory-preserving time integration methods. As such, the trajectory errors of particles were analyzed showing that once discontinuities are introduced to the flow, the proposed WENO scheme reduces the trajectory errors by more than an order of magnitude in comparison with different centered schemes. Present work introduces a high-order WENO-5 interpolation scheme. As discussed above, the focus here is solely on the interpolation methods, thus we consider only the interpolated quantities of interest such as velocity, pressure, temperature, etc., which are independent of the time-integrator temporal numerical errors. Hence, the results presented in this paper are applicable to essentially any time-integration method. The numerical tests in this paper focus on massless particles, yet the proposed methods and the results are applicable to particles with mass as well. Various extensive numerical tests are performed, including two-dimensional vortex flows, and realistic flows of interest such a normal shock wave, a three-dimensional spherical blast wave and a two-dimensional cellular detonation, which represents a typical highly-compressible reactive flow case. These test cases will demonstrate the advantages of WENO schemes over centered schemes and will shed some light on the source of the numerical oscillations for the high-order centered interpolation schemes.

2. Lagrangian tracking method

The equation of motion for a massless Lagrangian point particle is based on the following simple kinematic relation:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v} = \mathbf{u}\big(\mathbf{x}(t), t\big), \tag{1}$$

where $\mathbf{x}$ is the particle position, $\mathbf{v}$ is the particle velocity, and $\mathbf{u}$ is the flow velocity in the Eulerian frame of reference. Equation (1) can be integrated via the following initial condition for the particle position:

$$\mathbf{x}(t_0) = \mathbf{x}_0, \tag{2}$$

where t_0 is the initial time, and $\mathbf{x}_0$ is the initial particle position. Numerical integration of Eq. (1) requires the Eulerian velocity at the particle position. The Eulerian flow field, however, is typically known only at discrete points. As such, interpolation must be carried out in order to reconstruct the velocity at the particle position. In fact, this is required for any flow quantity of interest, such as pressure, temperature, density, etc. Also, the time derivative needs to be discretized by an integration method. Thus, solution of Eq. (1) involves both spatial and temporal numerical errors. In order to guarantee that the particle will follow the flow trajectory, these errors should be minimized as much as possible. It should be stressed that for massless particles, only the particle position depends both on the temporal discretization and spatial interpolation errors. All other flow quantities of interest, such as velocity, pressure, temperature, density, mass fraction of species, etc., depend solely on the spatial interpolation error. Note that the framework used in this paper is often referred to as an *online* approach. Hence, Eq. (1) is solved at each time step concurrently with the hydrodynamic integration after the flow field is evolved. This is in contrast with an *offline* approach, in which Eq. (1) is solved only after the flow simulation has ended. Yet, all the methods described in the current work are applicable to both approaches.

For the time integration, we use the Crank-Nicolson scheme:

$$\mathbf{x}^{n+1} = \mathbf{x}^n + \frac{1}{2}\Delta t\left(\mathbf{v}^n + \mathbf{v}^{n+1}\right) \tag{3}$$

The full integration procedure is described in [48]. We emphasize, though, that the focus of this current work is not time integration. All tests presented here focus on the interpolation of the flow quantities independently of the temporal discretization error. Thus, results presented in this paper are applicable to essentially any time integration method.

Different interpolation schemes are used for the spatial reconstruction of various quantities from a Eulerian grid to the particle position. These interpolation schemes can be represented by the weight functions [49]. The weight function, W, depends on the following single parameter that represents a dimensionless distance:

$$\tilde{x}_i = \frac{\left|x - x_{c_i}\right|}{\Delta x}, \tag{4}$$

where x is the particle x-position, x_{c_i} is the cell center x-position, and Δx is the grid spacing. Similar dimensionless parameters can be derived for the y- and z-directions. A quantity F can be interpolated from the cell centers to a particle position via the following formula:

$$F(x) = \sum_i W\left(\tilde{x}_i\right) F\left(x_{c_i}\right), \tag{5}$$

where the sum is taken over all adjacent cell centers, and the maximum value of i depends on the interpolation stencil size. The role of the weight function W in Eq. (5) is to prescribe the interpolation weights for each cell center according to its dimensionless distance from the particle position. All weight functions satisfy the following constraint:

$$\sum_i W\left(\tilde{x}_i\right) = 1. \tag{6}$$

This constraint guarantees that the weight function conserves the interpolated property [50]. Although the weight function is inherently one-dimensional (1D), it allows multidimensional interpolation [50], for instance, for the three-dimensional (3D) interpolation:

$$F(\mathbf{x}) = \sum_k \sum_j \sum_i W\left(\tilde{\mathbf{x}}_{i,j,k}\right) F\left(\mathbf{x}_{c_{i,j,k}}\right), \tag{7}$$

where

$$W\left(\tilde{\mathbf{x}}_{i,j,k}\right) = W\left(\frac{\left|x - x_{c_i}\right|}{\Delta x}\right) W\left(\frac{\left|y - y_{c_j}\right|}{\Delta y}\right) W\left(\frac{\left|z - z_{c_k}\right|}{\Delta z}\right), \tag{8}$$

and j and k are indices of adjacent cells in the y- and z-direction, respectively. One should note that an equally-spaced grid in each direction is assumed. It is important to mention that even for a multidimensional interpolation, the sum of weights still equals unity in a similar manner to Eq. (6), hence:

$$\sum_k \sum_j \sum_i W\left(\tilde{\mathbf{x}}_{i,j,k}\right) = 1 \tag{9}$$

Although this interpolation method is not multidimensional, as it is based on the product of one-dimensional weight functions, it has been successfully used for different applications [50–53]. A fully multidimensional interpolation can provide a higher level of accuracy. At the same time, the current approach is more computationally efficient [50], easier for implementation, and can provide the same order of accuracy as a fully multidimensional interpolation scheme with the same stencil size.

The following weight functions were implemented and investigated in the current work:

CIC (*Cloud-In-Cell*) [50]:

$$W\left(\tilde{x}\right) = \begin{cases} 1 - \tilde{x}, & \tilde{x} \le 1; \\ 0, & \tilde{x} > 1. \end{cases} \tag{10}$$

This scheme is a linear interpolation, and as such it has a 2-point stencil and it is 2nd-order accurate. It should be noted that its weights are continuous and positive.

TSC (*Triangular-Shaped-Cloud*) [50]:

$$W\left(\tilde{x}\right) = \begin{cases} \frac{3}{4} - \tilde{x}^2, & \tilde{x} < 0.5; \\ \frac{1}{2}\left(\frac{3}{2} - \tilde{x}\right)^2, & 0.5 \le \tilde{x} \le 1.5; \\ 0, & \tilde{x} > 1.5. \end{cases} \tag{11}$$

This scheme can be derived with a quadratic polynomial B-spline, and it has a 3-point stencil. As CIC, its order of accuracy is 2nd, and its weights are also strictly positive. However, it has several advantages over CIC. First of all, both weight values and the 1st derivative are continuous. Second, the locality errors associated with the interpolation position are greatly reduced in comparison with CIC [50]. These properties made this scheme useful for different applications, such as plasma physics and astrophysical flows [51–53].

QP (*Quadratic Polynomial*) [52]:

$$W\left(\tilde{x}\right) = \begin{cases} 1 - \tilde{x}^2, & \tilde{x} < 0.5; \\ \frac{1}{2}\left(1 - \tilde{x}\right)\left(2 - \tilde{x}\right), & 0.5 \le \tilde{x} \le 1.5; \\ 0, & \tilde{x} > 1.5. \end{cases} \tag{12}$$

This scheme is derived using quadratic Lagrange polynomials, and as such it provides 3rd order of accuracy with a 3-point stencil. Although its order of accuracy is greater than that of TSC with the same stencil size, its weight values are discontinuous, which means it is more oscillatory [52].

W_4 [54]:

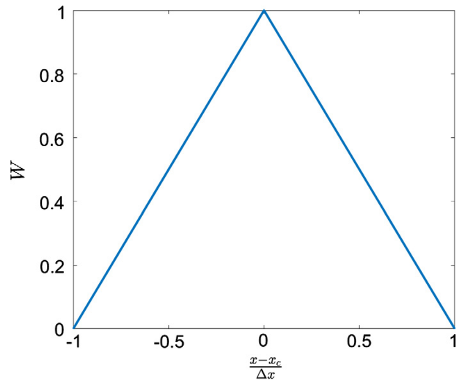
$$W\left(\tilde{x}\right) = \begin{cases} 1 - \frac{5}{2}\tilde{x}^2 + \frac{3}{2}\tilde{x}^3, & \tilde{x} < 1; \\ \frac{1}{2}\left(2 - \tilde{x}\right)^2\left(1 - \tilde{x}\right), & 1 \le \tilde{x} \le 2; \\ 0, & \tilde{x} > 2. \end{cases} \tag{13}$$

This scheme is based on a cubic B-spline, and as such it has a 4-point stencil. Similar to QP, it is 3rd-order accurate, however, similarly to TSC, its weight values and the 1st derivative are continuous.

Everett's interpolation formula [54]:

$$W\left(\tilde{x}\right) = \begin{cases} \frac{1}{2}\left(2 - \tilde{x}\right)\left(1 - \tilde{x}^2\right), & \tilde{x} < 1; \\ \frac{1}{6}\left(2 - \tilde{x}\right)\left(3 - \tilde{x}\right)\left(1 - \tilde{x}\right), & 1 \le \tilde{x} \le 2; \\ 0, & \tilde{x} > 2. \end{cases} \tag{14}$$

This scheme can be derived with a cubic polynomial, and similarly to W_4, it has a 4-point stencil. It is, however, 4th-order accurate and only its weight values are continuous.



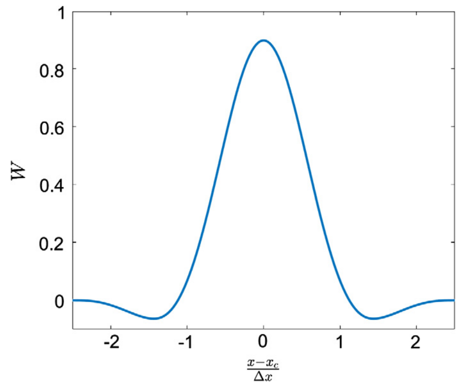


Fig. 1. Two weight functions, showing weight against the dimensionless distance: (a) CIC. (b) W_5.

W_5 [54]:

$$W\left(\tilde{x}\right)=\begin{cases}\frac{1}{48}\left(\frac{345}{8}-75\tilde{x}^2+42\tilde{x}^4\right), & \tilde{x}<\frac{1}{2};\\ \frac{1}{48}\left(\frac{165}{4}+20\tilde{x}-150\tilde{x}^2+120\tilde{x}^3-28\tilde{x}^4\right), & \frac{1}{2}\le\tilde{x}\le\frac{3}{2};\\ \frac{1}{48}\left(\tilde{x}-\frac{5}{2}\right)^3\left(7\tilde{x}-\frac{15}{2}\right), & \frac{3}{2}<\tilde{x}\le\frac{5}{2};\\ 0, & \tilde{x}>\frac{5}{2}.\end{cases} \tag{15}$$

This scheme is an improved version of W_4 with a 5-point stencil. It can be derived with a quartic B-spline. It is 4th-order accurate, as Everett's interpolation formula, yet the weight values and their first two derivatives are continuous.

Note that for all weight functions with the order of accuracy greater than 2nd weights become negative [49]. Fig. 1 illustrates this point by showing two weight functions, CIC and W_5. The weights of CIC as a function of the dimensionless distance, see Eq. (4), can be seen in Fig. 1a. The weight function has a triangular shape, and all the weight values vary between zero and unity. As CIC is only 2nd order accurate, its weights can be strictly positive. In contrast, Fig. 1b shows the weights of W_5 as a function of the dimensionless distance. As the order of accuracy for W_5 is 4th, weights attain negative values. As will be shown later, the negative weights induce numerical oscillations once discontinuous values are being interpolated.

We present two interpolation schemes WENO-3 and WENO-5 [55] that remedy this issue. As these interpolation schemes are not centered, the weight function depends on the following, slightly altered, dimensionless parameter:

$$\tilde{x}=\frac{x-x_{c_i}}{\Delta x}. \tag{16}$$

WENO-3:

This interpolation scheme is derived using 2nd-order Lagrange polynomials, which results in a 3-point stencil. Once the quantities being interpolated are sufficiently smooth, it becomes identical to the QP centered scheme described above. Thus, as long as the flow gradients are low, the scheme is 3rd-order accurate. At the same time, once discontinuities, like shocks, are introduced into the flow, the scheme is reduced to a 2nd-order linear interpolation or extrapolation. The scheme is comprised of two sub-stencils, with each sub-stencil being associated with a linear weight for the left and right directions [55]:

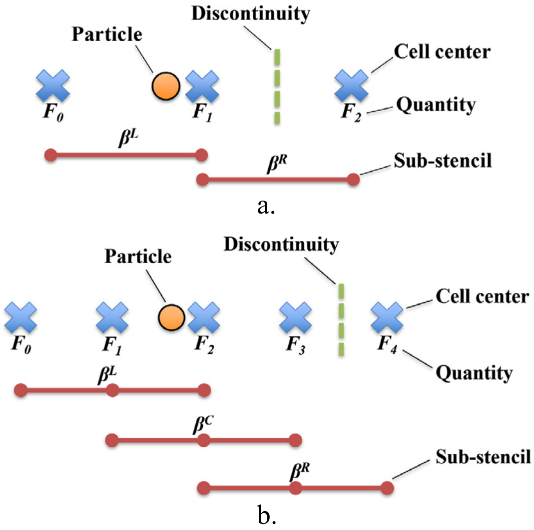


Fig. 2. Schematic description of the WENO interpolation procedure: (a) WENO-3. (b) WENO-5.

$$C^L = \frac{1}{2}\left(1-\tilde{x}\right), \qquad C^R = \frac{1}{2}\left(1+\tilde{x}\right). \tag{17}$$

Each sub-stencil has its own smoothness indicator, which provides an estimate for the gradients of the interpolated quantity in the sub-stencil:

$$\beta^L_{j,k} = \left(F_{1,j,k} - F_{0,j,k}\right)^2, \qquad \beta^R_{j,k} = \left(F_{2,j,k} - F_{1,j,k}\right)^2, \tag{18}$$

where j and k are indices for the y- and z-directions, and their values vary based on the stencil size. In this case, for a 3-point stencil, their range is 0–2. F can be any cell-centered flow quantity, e.g., velocity, pressure, temperature, etc. The non-linear weights are calculated via the following auxiliary variables:

$$\alpha^L_{j,k} = \frac{C^L}{\left(\beta^L_{j,k} + \varepsilon\right)^2}, \qquad \alpha^R_{j,k} = \frac{C^R}{\left(\beta^R_{j,k} + \varepsilon\right)^2}, \tag{19}$$

where ε is a small value in order to avoid division by zero, which is chosen as 10^{-14}. One may note that the values of ε suggested in the literature vary significantly from 10^{-6} to as low as 10^{-40} [55–57]. The value chosen proved to be sufficiently small for all the tests performed in the current work.

The resulting non-linear weights for the two sub-stencils are given by the following relations:

$$\omega^L_{j,k} = \frac{\alpha^L_{j,k}}{\alpha^R_{j,k} + \alpha^L_{j,k}}, \qquad \omega^R_{j,k} = \frac{\alpha^R_{j,k}}{\alpha^R_{j,k} + \alpha^L_{j,k}}. \tag{20}$$

Finally, the interpolation scheme can be presented in the form of the following weight function:

$$W_{j,k}\left(\tilde{x}\right) = \begin{cases} 0, & \tilde{x} < -\frac{3}{2}; \\ -\omega^L_{j,k}\tilde{x}, & -\frac{3}{2} \le \tilde{x} \le -\frac{1}{2}; \\ \omega^L_{j,k}\left(1+\tilde{x}\right) + \omega^R_{j,k}\left(1-\tilde{x}\right), & -\frac{1}{2} < \tilde{x} \le \frac{1}{2}; \\ \omega^R_{j,k}\tilde{x}, & \frac{1}{2} < \tilde{x} \le \frac{3}{2}; \\ 0, & \tilde{x} > \frac{3}{2}. \end{cases} \tag{21}$$

The typical representation of the interpolation scheme can be found in [55], though it should be noted that the weight function representation derived here is exactly equivalent. This new form is useful for the implementation in the code, as it has the same structure as the classical centered weight functions.

Note that in contrast to the centered schemes described above, the weight function (21) is asymmetric. Furthermore, its weights are non-linear, as they depend not only on the particle position but also on the cell-centered quantity gradients. Thus, the weights of the weight function (21) adapt themselves in order to avoid interpolation across a discontinuity. A schematic that depicts the procedure for the WENO-3 interpolation scheme is shown in Fig. 2a. For the presented case, the discontinuity passes through the right sub-stencil, thus, the value of β^R will be large. As such, the non-linear weight for the right sub-stencil will be close to zero. Thus, interpolation will be performed almost solely with the left sub-stencil, where the flow is continuous, and interpolation through the discontinuity will be avoided.

Multidimensional interpolation can be carried out in a way similar to Eq. (7) for the centered schemes. For WENO schemes, however, even for an equally spaced grid, the one-dimensional weights at different positions are not identical because the flow gradients can be different. Thus, for a 3D interpolation, the following formula is used:

$$F\left(\mathbf{x}\right)=\sum_{k}\sum_{j}\sum_{i}W_{i,j,k}\left(\tilde{\mathbf{x}}\right)F\left(\mathbf{x}_{c_{i,j,k}}\right), \tag{22}$$

where

$$W_{i,j,k}\left(\tilde{\mathbf{x}}\right)=W_{j,k}\left(\frac{x-x_{c_i}}{\Delta x}\right)W_k\left(\frac{y-y_{c_j}}{\Delta y}\right)W\left(\frac{z-z_{c_k}}{\Delta z}\right). \tag{23}$$

This formulation is based on a dimension-by-dimension approach. Note that the x-direction weights are now two-dimensional and depend also on the y- and z-coordinates, and the y-direction weights are now one-dimensional and depend also on the z-coordinate. The choice of the order of the dimensional sweeps is arbitrary, and although a different choice can change the results, they should not be significantly different [58]. The terms W_k and W in Eq. (23) are weights along the centerline of the stencil. They can be calculated via Eq. (21) by setting $j=1$ for W_k, and $j=1$, $k=1$ for W.

WENO-5:

This scheme is derived using 4th-order Lagrange polynomials. It is based on a 5-point stencil and it is 5th-order accurate for smooth flows with the accuracy dropping to a minimum of 3rd-order in the vicinity of shocks [55]. The interpolation scheme is divided into three sub-stencils associated with the following linear weights for the left, center, and right positions [55]:

$$C^L=\frac{1}{12}\left(\tilde{x}-1\right)\left(\tilde{x}-2\right),\qquad C^C=-\frac{1}{6}\left(\tilde{x}+2\right)\left(\tilde{x}-2\right),\qquad C^R=\frac{1}{12}\left(\tilde{x}+2\right)\left(\tilde{x}+1\right). \tag{24}$$

The smoothness indicators are defined as follows:

$$\beta^L_{j,k}=\frac{13}{12}\left(F_{0,j,k}-2F_{1,j,k}+F_{2,j,k}\right)^2+\frac{1}{4}\left(F_{0,j,k}-4F_{1,j,k}+3F_{2,j,k}\right)^2, \tag{25}$$

$$\beta^C_{j,k}=\frac{13}{12}\left(F_{1,j,k}-2F_{2,j,k}+F_{3,j,k}\right)^2+\frac{1}{4}\left(F_{1,j,k}-F_{3,j,k}\right)^2, \tag{26}$$

$$\beta^R_{j,k}=\frac{13}{12}\left(F_{2,j,k}-2F_{3,j,k}+F_{4,j,k}\right)^2+\frac{1}{4}\left(3F_{2,j,k}-4F_{3,j,k}+F_{4,j,k}\right)^2, \tag{27}$$

where j and k are indices for the y- and z-directions varying according to the stencil size, in this case in the range of 0–4. The same auxiliary variables described in Eq. (19) for the left and right positions are used here. Also, the following auxiliary variable is calculated for the center position

$$\alpha^C_{j,k}=\frac{C^C}{\left(\beta^C_{j,k}+\varepsilon\right)^2}. \tag{28}$$

Accordingly, the non-linear weights for the three sub-stencils are calculated as follows:

$$\omega^L_{j,k}=\frac{\alpha^L_{j,k}}{\alpha^L_{j,k}+\alpha^C_{j,k}+\alpha^R_{j,k}},\qquad \omega^C_{j,k}=\frac{\alpha^C_{j,k}}{\alpha^L_{j,k}+\alpha^C_{j,k}+\alpha^R_{j,k}},\qquad \omega^R_{j,k}=\frac{\alpha^R_{j,k}}{\alpha^L_{j,k}+\alpha^C_{j,k}+\alpha^R_{j,k}}. \tag{29}$$

Finally, the weight function for the WENO-5 interpolation scheme is defined as follows:

$$W_{j,k}\left(\tilde{x}\right)=\begin{cases}0, & \tilde{x}<-\frac{5}{2};\\ \frac{1}{2}\left(\tilde{x}+1\right)\tilde{x}\omega^L_{j,k}, & -\frac{5}{2}\le\tilde{x}<-\frac{3}{2};\\ -\left(\tilde{x}+2\right)\tilde{x}\omega^L_{j,k}+\frac{1}{2}\tilde{x}\left(\tilde{x}-1\right)\omega^C_{j,k}, & -\frac{3}{2}\le\tilde{x}<-\frac{1}{2};\\ \frac{1}{2}\left(\tilde{x}+2\right)\left(\tilde{x}+1\right)\omega^L_{j,k}-\left(\tilde{x}+1\right)\left(\tilde{x}-1\right)\omega^C_{j,k}+\frac{1}{2}\left(\tilde{x}-1\right)\left(\tilde{x}-2\right)\omega^R_{j,k}, & -\frac{1}{2}\le\tilde{x}\le\frac{1}{2};\\ \frac{1}{2}\left(\tilde{x}+1\right)\tilde{x}\omega^C_{j,k}-\tilde{x}\left(\tilde{x}-2\right)\omega^R_{j,k}, & \frac{1}{2}<\tilde{x}\le\frac{3}{2};\\ \frac{1}{2}\tilde{x}\left(\tilde{x}-1\right)\omega^R_{j,k}, & \frac{3}{2}<\tilde{x}\le\frac{5}{2};\\ 0, & \tilde{x}>\frac{5}{2}.\end{cases}. \tag{30}$$

Again, we refer the reader to [55] for the traditional form of this interpolation scheme.

Fig. 2b shows a schematic that depicts the procedure for the WENO-5 interpolation scheme. For the presented case, the discontinuity passes through the right sub-stencil, hence, its weight will be essentially zero. Thus, interpolation will be performed using the left and center sub-stencils, dropping the interpolation order of accuracy to 4th.

As mentioned above, multidimensional interpolation is possible as described by Eqs. (22)–(23), and the terms W_k and W in Eq. (23) are weights along the centerline of the stencil. They can be calculated via Eq. (21), in this case, by setting $j = 2$ for W_k, and $j = 2$, $k = 2$ for W. Note that similarly to the centered schemes, Eqs. (6), (9) hold for both WENO-3 and WENO-5. Thus, the sum of weights for WENO schemes always equals unity and the interpolated quantity is fully conserved. Also, for the multidimensional interpolation, WENO schemes can handle accurately multiple discontinuities in the stencil, as long as they do not cross all sub-stencils of any 1D weight function. Note that, similarly to the treatment for centered schemes discussed above, the multidimensional interpolation for WENO schemes is based on the product of the one-dimensional weight functions. Thus, it is not fully multidimensional, and it is performed in a dimension-by-dimension fashion. This approach has been commonly used for both interpolation and reconstruction [59], and it is considered highly efficient for Cartesian structured grids [45], which are utilized in the current work. Note that fully multidimensional WENO reconstruction is also possible [58,60,61], and it can provide a higher level of accuracy, however, the order of accuracy remains the same [60], and the implementation is much more involved [59] with an additional computational cost. For unstructured grids the dimension-by-dimension approach cannot be used, and different fully multidimensional formulations have been developed, including for 2D triangular [62,63] and 3D tetrahedral meshes [64], as well as for adaptive mesh refinement [65] and moving meshes [66].

3. Results and discussion

3.1. Steady-state non-reactive flows

The first two test cases involve steady-state conditions, in which the flow field is stationary, and thus the flow equations are not solved.

3.1.1. Two-dimensional vortex

The first test case is a 2D steady-state vortex flow. Two different cases are compared, a vortex with a uniform and continuous flow (Fig. 3a), and a shocked vortex with an abrupt velocity magnitude jump in the middle of the domain (Fig. 3b). The flow field for both cases can be described by the following relations:

$$u = \frac{dx}{dt} = -u_\infty \sin(\theta), \qquad v = \frac{dy}{dt} = u_\infty \cos(\theta), \tag{31}$$

where u and v are the x- and y-velocity components, and u_∞ is the velocity magnitude. The corresponding cylindrical radial and tangential components, r and θ, can be defined as follows:

$$r = \sqrt{(x - x_c)^2 + (y - y_c)^2}, \qquad \theta = \operatorname{atan2}(y - y_c, x - x_c), \tag{32}$$

where x_c and y_c are the vortex center coordinates, and $\operatorname{atan2}(y, x)$ is defined as follows:

$$\operatorname{atan2}(y, x) = \begin{cases} \arctan(y/x), & x > 0; \\ \arctan(y/x) + \pi, & x < 0;\ y \geq 0; \\ \arctan(y/x) - \pi, & x < 0;\ y < 0; \\ \pi/2, & x = 0;\ y > 0; \\ -\pi/2, & x = 0;\ y < 0; \\ \text{undefined}, & x = 0;\ y = 0. \end{cases} \tag{33}$$

For a uniform flow vortex, u_∞ is constant, whereas for the shocked case, it depends on the x-coordinate according to the following equation:

$$u_\infty(x) = \begin{cases} u_h, & x \leq x_c; \\ u_l, & x > x_c, \end{cases} \tag{34}$$

where u_h and u_l are high and low velocity magnitudes. Note that the shocked vortex test case does not represent a realistic flow configuration. It does allow us, however, to demonstrate the numerical errors that arise once discontinuities in the flow are introduced in comparison with the uniform case. More realistic shocked flow configurations will be discussed in the following test cases.

The problem is solved on a 1×1 m square domain with the periodic boundary conditions. The vortex center is located at $x_c = y_c = 0.5$ m. For the continuous case, the value of u_∞ is set to 100π m/s. For the shocked case, the following values

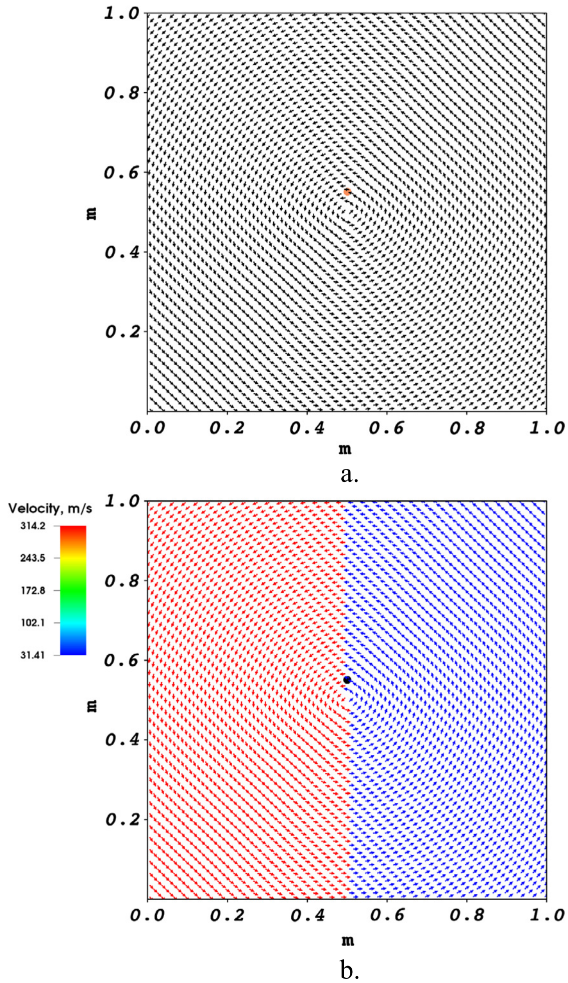


Fig. 3. Velocity vector flow field for the two-dimensional vortex problem. The colored dot marks the initial position of the Lagrangian particle: (a) Uniform flow. (b) Shocked flow. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

are set: $u_h = 100\pi$ m/s and $u_l = 10\pi$ m/s. These values correspond to the typical order of magnitude in the velocity jump across strong shocks or flame fronts. The time-step is governed by the particle CFL number that can be defined as follows:

$$\mathrm{CFL} = \frac{u_{\max}\Delta t}{\Delta x}, \tag{35}$$

where $u_{\max}$ is the particle maximum possible velocity, Δt is the time-step size, and Δx is the grid spacing. For the two cases above, $u_{\max} = u_\infty = u_h$, hence, the definition for the CFL number is identical for both of them.

First, we can use the uniform flow vortex configuration to verify the order of accuracy for different interpolation schemes. The accuracy is tested by seeding a single particle at a radius of 0.05 m from the vortex center (see Fig. 3a), and propagating the particle through the flow for 1 ms. Velocity magnitude is interpolated to the particle position, and the L_∞ norm of the error is calculated for each of the interpolation schemes that were presented above. Four different grid sizes are used: 100 $\times$ 100, 200 $\times$ 200, 400 $\times$ 400, 800 $\times$ 800 with a small CFL number of about 5×10^{-4}. Note that for massless particles, the velocity error depends only on the accuracy of spatial interpolation as only the particle position propagates in time, see Eq. (1). Thus, this test can be performed with any CFL number or time integration method as long as the particle retains its circular trajectory. As Crank-Nicolson is trajectory preserving, the radial position error is bounded in time and thus negligible [48]. Fig. 4 shows the L_∞ error of the velocity magnitude as a function of grid spacing, Δx, for different interpolation schemes. The velocity errors for all schemes show the expected order of spatial accuracy as described above. One may note that QP and WENO-3 give the same results, as for smooth flows WENO-3 reduces to QP. Also, for the current test case, the interpolation errors for several interpolation methods are identical to within several digits after the decimal point. Thus, the corresponding curves overlap. In the current figure, CIC is hidden by TSC, QP is hidden by WENO-3, and Everett's interpolation formula is hidden by W_5.

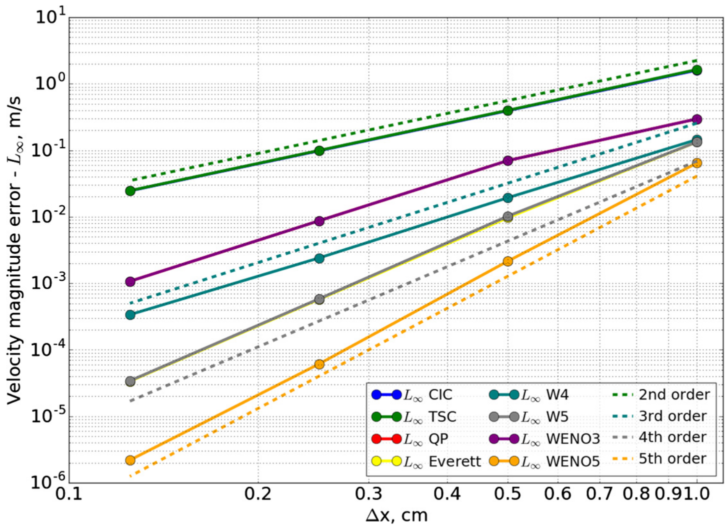


Fig. 4. Spatial accuracy test. L_∞ velocity magnitude errors as a function of the grid spacing Δx. CIC is hidden by TSC, QP is hidden by WENO-3, and Everett's interpolation formula is hidden by W_5.

Table 1
Comparison of the interpolation schemes for the uniform and shocked flow vortices. L_∞ normalized velocity magnitude errors.

Weight function	2-D vortex Normalized velocity error – L_∞	2-D vortex – shock Normalized velocity error – L_∞
CIC	5.112×10^{-3}	4.490
TSC	5.213×10^{-3}	4.491
QP	9.462×10^{-4}	4.465
Everett	4.249×10^{-4}	4.509
W_4	4.614×10^{-4}	4.509
W_5	4.249×10^{-4}	4.509
WENO-3	9.462×10^{-4}	1.374×10^{-2}
WENO-5	2.057×10^{-4}	3.900×10^{-3}

The second test involves a comparison of the relative errors for the two flow configurations and all interpolation schemes. The relative error, E_u, is given by the following relation:

$$E_u = \frac{|U - u_{exact}|}{u_{exact}}, \tag{36}$$

where U and u_{exact} are the interpolated and exact velocity magnitudes, respectively, with the velocity magnitude defined as follows:

$$U = \sqrt{u^2 + v^2}. \tag{37}$$

The test is performed on a coarse grid of 100 × 100 cells, CFL number of about 5×10^{-4}, and the entire simulation proceeds until the particle completes 5 orbits. Table 1 shows the L_∞ relative velocity magnitude error for all interpolation schemes, and the two flow configurations. One can see that for the uniform flow vortex, the errors for all schemes are lower than 1%. Yet, the errors for CIC and TSC are the largest, whereas the errors for QP and WENO-3 are much smaller, and for other schemes, the errors are about an order of magnitude smaller. For the shocked vortex case, the errors are significant for all centered schemes with values well above 100%. On the other hand, WENO schemes show significantly smaller errors: WENO-3 has an error of about 1% and WENO-5 of about 0.4%.

Tests presented above confirm quantitatively that the accuracy of WENO schemes is comparable to centered schemes for smooth flows. Once sharp discontinuities are introduced to the flow, centered schemes fail and WENO schemes are proven much more accurate. The next test case will demonstrate the advantage of WENO schemes over centered schemes for a more realistic flow configuration, including interpolation of different flow quantities of interest. Also, this test will shed light on the nature of the interpolation errors in centered schemes.

3.1.2. Normal shock wave

For this test case, the classical steady state normal shock solution is being investigated. The domain size is 1 × 0.25 m with a grid of 320 × 80 cells. The flow is 1D in the horizontal direction and the shock is located at 0.5 m. The particle is seeded before the shock and is advected by the flow until it passes the shock. Upstream conditions correspond to pressure, temperature and Mach number of 5×10^4 Pa, 250 K, and 10, respectively. The gas is assumed to be air with a calorically perfect equation of state. The heat capacity ratio and specific gas constant are 1.4 and 287 J/(kg K), respectively. Accordingly, downstream conditions are calculated using normal shock relations [67]. The particle is integrated with the flow CFL number, which is 0.45, and defined as follows:

$$\mathrm{CFL} = \frac{(|V_{\max}| + a)\,\Delta t}{\Delta x} \tag{38}$$

where $V_{\max}$ is the maximum flow velocity and a is the local speed of sound. Three flow quantities of interest are interpolated to the particle position: pressure, temperature, and density. These interpolated quantities are typically of primary interest for the Lagrangian analysis of different flows. Yet, the possible applications can be much broader than that. For example, in simulations of multi-phase reacting flows, these quantities would determine the heat, and mass transfer rates for a fuel spray droplet represented as a point Lagrangian particle [68,69]. For the current test case, it is possible to compare the interpolated values at each position with the exact solution, as the flow field is stationary and known through simple analytic relations.

Fig. 5a shows pressure as a function of position in the vicinity of the shock wave. The exact solution is compared with values interpolated using centered schemes, CIC and W_5, and the two WENO schemes. One can see that for the exact solution, pressure increases sharply at 0.5 m. However, for CIC, the pressure rise is not sharp and the shock is smeared. For W_5, not only is the shock not sharp, but the interpolated values undershoot and overshoot, before and after the shock, respectively. Note that the undershoot is so large that the absolute pressure attains nonphysical negative values. Thus, for calculations with various physical processes dependent on pressure or temperature, e.g., mass transfer or reaction rates, these negative values can lead to a complete breakdown of the calculation. For both WENO schemes, which are effectively indistinguishable for the current case, the pressure rise is much sharper than for the centered schemes, and there are no visible oscillations.

As mentioned above, since CIC has strictly positive weights, it does not oscillate. In contrast, W_5 has an order of accuracy greater than 2nd, hence some of its weights must be negative, and oscillations can occur. The numerical oscillations can be explained through the current test case. As the particle approaches the shock, both high- and low-pressure values are interpolated to the particle position. The high-pressure values receive negative weights. Although these weight values are low, the multiplication with the high-pressure values makes them dominant in comparison with the interpolated low-pressure values. The result is a nonphysical negative pressure. Once the particle passes the shock, low-pressure values receive negative weights. In this case, the contribution of these negative weights to the final result is negligible in comparison with the high-pressure values. However, as the sum of all weights always equals unity, the sum of the high-pressure weights must be greater than one. As a result, the interpolated high pressure is augmented above its correct value and an overshoot occurs. Note that grid refinement can reduce the smearing of the shock, although it may require excessively high resolution that may not be practical. Yet, the oscillations cannot be avoided in this way. In fact, their magnitude is independent of the grid spacing, and they will just shift closer to the discontinuity. On the other hand, for WENO schemes, the oscillation magnitude decreases with the grid size [45]. It is important to stress that these trends are completely general for any centered weight function with no special treatment. For instance, TSC, which is 2nd order accurate and has only positive weights, will smear the shock in a manner similar to CIC. As its stencil is even wider than CIC, the smearing will occur over 3 cells rather than 2. Finally, for any other high order interpolation scheme besides W_5, oscillations will also occur, as the negative weights are unavoidable as mentioned above. Only their magnitude and exact position relative to the discontinuity will vary.

Fig. 5b shows a comparison between the exact and interpolated temperatures as a function of position. The trends are very similar to Fig. 5a, as both CIC and W_5 smear the temperature profile and WENO schemes capture it in a sharper manner. For W_5, oscillations also occur, though, temperature values at the undershoot are not negative, but they reach extremely low values ($\sim$10 K). This is a numerical artifact of the interpolation as values lower than the initial temperature are not possible for this problem according to the second law of thermodynamics [67].

Fig. 5c shows the same plot for density. Again, the same trends can be observed. For W_5, the magnitude of the oscillations is much lower than for pressure and temperature described above. This is related to the fact that the density jump across a shock is bounded in the limit of high Mach numbers, unlike pressure and temperature. This makes the oscillations less severe, yet even so, for the undershoot, the maximum relative interpolation error is higher than 20% and the interpolated values are inconsistent with the exact solution.

This test demonstrates the advantages of WENO schemes in comparison with centered schemes for a basic flow configuration. However, this test case describes a steady state 1D flow with a sharp discontinuity. Realistic flows are 3D and time-dependent. Also, many flow solvers are based on shock capturing schemes, which cause the shock to get smeared across several grid cells [70]. The next test case will show that the same trends observed for a simple normal shock are retained for a much more complex shocked flow configuration.

3.2. Unsteady non-reactive flow

Three-dimensional spherical blast wave

This test case is similar to a strong blast wave problem presented in [71]. The current test case, however, is 3D and does not involve magnetohydrodynamics. Fig. 6 shows the initial setup of the problem. The domain size is $1 \times 1 \times 1$ m

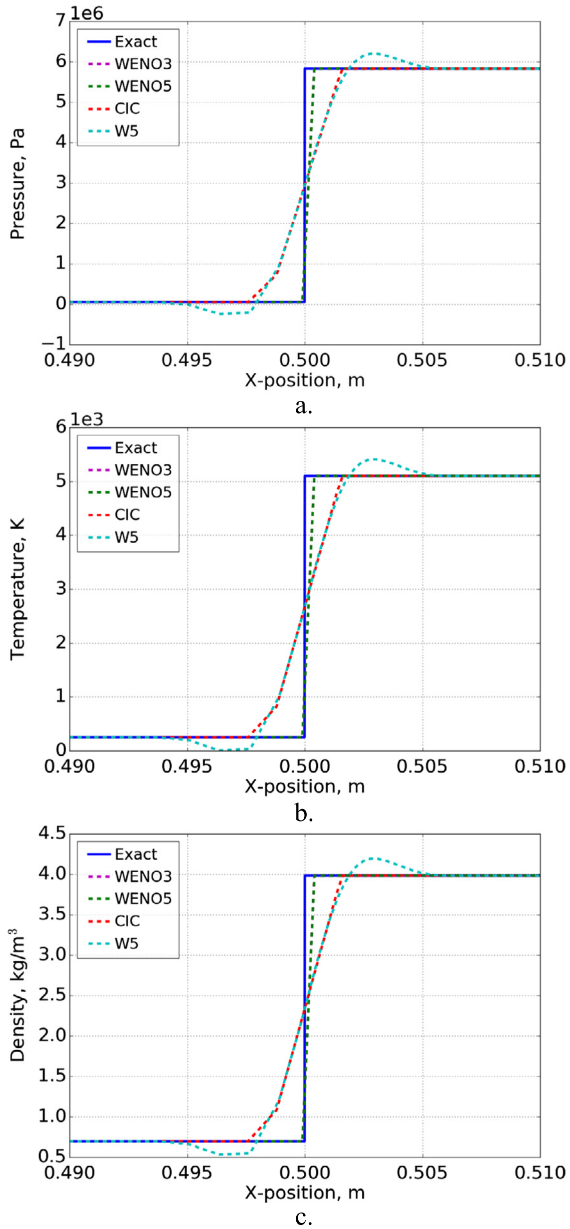


Fig. 5. Normal shock test. Comparison between the exact solution, and the interpolated results for CIC, W_5, WENO-3 and WENO-5: (a) Pressure. (b) Temperature. (c) Density.

with 200 × 200 × 200 grid cells and periodic boundary conditions. As for the normal shock test case, the gas is assumed to be calorically perfect with heat capacity ratio and specific gas constant of 1.4 and 287 J/(kg K), respectively. Initially, the entire domain has pressure of 1 atm, density of 1.2 kg/m^3 and zero velocity, except for a sphere with a radius of 0.1 m from the origin, where the pressure is initially set to 1000 atm. Also, a single particle is seeded at $x = y = z = 0.15$ m. The governing equations are the compressible Euler equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \tag{39}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla P = 0, \tag{40}$$

$$\frac{\partial E}{\partial t} + \nabla \cdot ((E + P)\,\mathbf{u}) = 0, \tag{41}$$

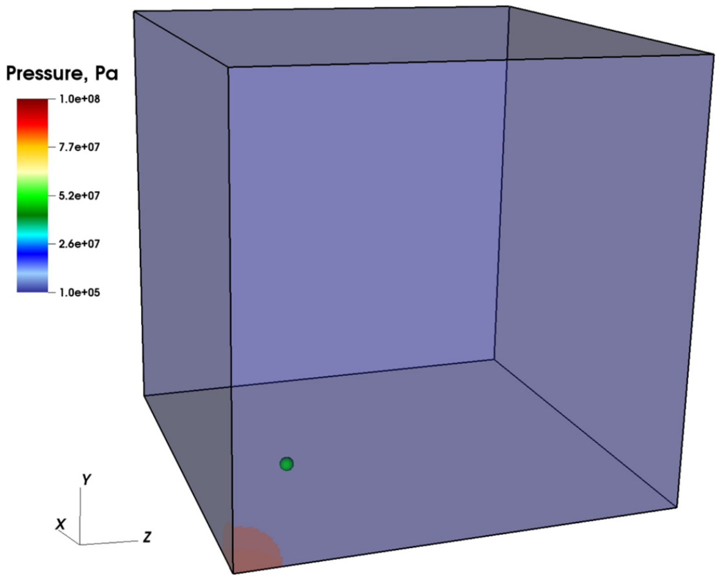


Fig. 6. Pressure distribution for the initial setup of the three-dimensional spherical blast wave test. The green sphere marks the initial position of the Lagrangian particle.

where ρ is density, $\mathbf{u}$ is velocity, P is pressure, and E is the energy density. Equations are solved using the fixed grid massively parallel code Athena-RFX [72], which is a reactive flow extension to the Athena magnetohydrodynamics code [73]. The flow equations integration algorithm is the unsplit corner transport upwind (CTU) [74], which is a fully conservative high-order Godunov-type method. The integration scheme utilizes piecewise parabolic method (PPM) [75] for spatial reconstruction and the HLLC approximate Riemann solver [42] for calculating fluxes at cell faces. The method is 3rd order accurate in space and 2nd order accurate in time. Particle integration is performed at the end of each time-step, as massless particle equations are one-way coupled to the flow equations. All the tests are performed with a flow CFL number of 0.45.

The initial flow field creates a strong spherical blast wave that passes through the point particle. Pressure, temperature, and density are interpolated to a particle position using CIC, W_5, and the two WENO schemes. An exact solution for the interpolated values on the Lagrangian trajectory cannot be derived, thus, comparison between different interpolation schemes is only qualitative. Also, note that even if an exact solution could have been derived, deviations from the analytical solution would arise both from the numerical errors associated with the flow solver and from the solution of the Lagrangian particle evolution equation, and it would be very difficult to determine the individual effect of each error. Fig. 7a shows a comparison of the interpolated pressure as a function of time for different interpolation schemes. In general, trends are very similar to the previous normal shock test case. Both CIC and W_5 smear the shock in comparison with the two WENO schemes that capture its structure in a sharper manner. Also, for W_5, the interpolated pressure undershoots to a nonphysically small value of $\sim$20 kPa. Fig. 7b shows the same plot for the interpolated temperature. Again, the trends are the same. Here for W_5 the temperature drops by about 50 K from its initial value of about 300 K. As discussed above for the normal shock case, this result is a numerical artifact that violates the second law of thermodynamics [67]. Fig. 7c shows results for the interpolated density. The capturing of the discontinuity is similar to the previous graphs, as CIC and W_5 tend to smear it, and WENO schemes reproduce it sharply. For W_5, there is an undershoot before the shock, although its magnitude is rather low. Similar to the previous test, density is bounded according to the conservation laws, and thus oscillations are reduced for this case.

It can be concluded that the trends observed for the normal shock case are present for this 3D time-dependent simulation, where the shocks are broadened over several grid cells. Thus, WENO interpolation schemes are beneficial even at these conditions.

3.3. Unsteady reactive flow

Two-dimensional detonation

An important example of a highly compressible, realistic reactive flow configuration is a multidimensional gas-phase detonation. For sufficiently high activation energies, detonations tend to be unstable with a complex interaction between transverse waves and triple-shock configurations that create cellular structures [76]. This phenomenon has been known for over half a century and has been studied both experimentally [77] and numerically [78,79]. Today, with the increase in computational power, numerical simulations of multidimensional detonations are quite common, including investigations of two- and three-dimensional configurations with single- and even multi-step chemical reaction kinetics [76,80–82]. In

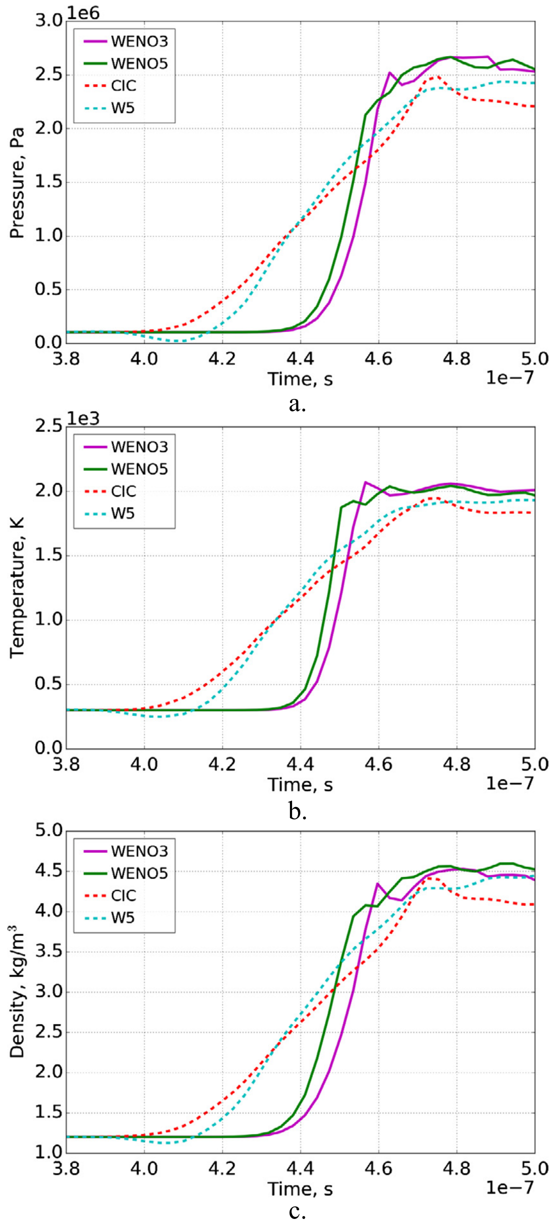


Fig. 7. Three-dimensional spherical blast wave test. Comparison between the interpolated results for CIC, W_5, WENO-3 and WENO-5: (a) Pressure. (b) Temperature. (c) Density.

order to demonstrate the benefits of the proposed Lagrangian-particle approach for this type of flows, a 2D problem with a single-step reaction kinetics is solved.

The governing equations for the problem are the compressible reactive Euler equations with the calorically perfect gas equation of state. The full set of equations includes Eqs. (39)–(40) and the following energy and species equations:

$$\frac{\partial E}{\partial t} + \nabla \cdot ((E + P)\,\mathbf{u}) = -\rho q \dot{w}, \tag{42}$$

$$\frac{\partial \rho Y}{\partial t} + \nabla \cdot (\rho Y \mathbf{u}) = \rho \dot{w}, \tag{43}$$

where q is the chemical energy release, $\dot{w}$ is the reaction source term, and Y is the mass fraction of the reactant. Note that for detonations, molecular transport effects like heat conduction, diffusive mass transfer or viscous shear stresses have typically a negligible effect on the solution. For chemical reactions, a first-order single-step Arrhenius kinetics is assumed:

Table 2
Input parameters and computed properties for the single-step reaction model.

P_0	1.01325×10^5 Pa	Initial pressure
T_0	293 K	Initial temperature
ρ_0	0.87345 kg/m^3	Initial density
γ	1.17	Adiabatic index
M	0.021 kg/mol	Molecular weight
A	3.56×10^7 m^3/(kg s)	Pre-exponential factor
Q	$24.199 \cdot RT_0$	Activation energy
q	$43.28 \cdot RT_0/M$	Chemical energy release
D_{CJ}	1.993×10^3 m/s	CJ detonation velocity
P_{ZND}	$31.47 \cdot P_0$	von Neumann pressure
T_{ZND}	$3.457 \cdot T_0$	von Neumann temperature
ρ_{ZND}	$9.104 \cdot \rho_0$	von Neumann density
x_d	1.927×10^{-4} m	Half-reaction thickness

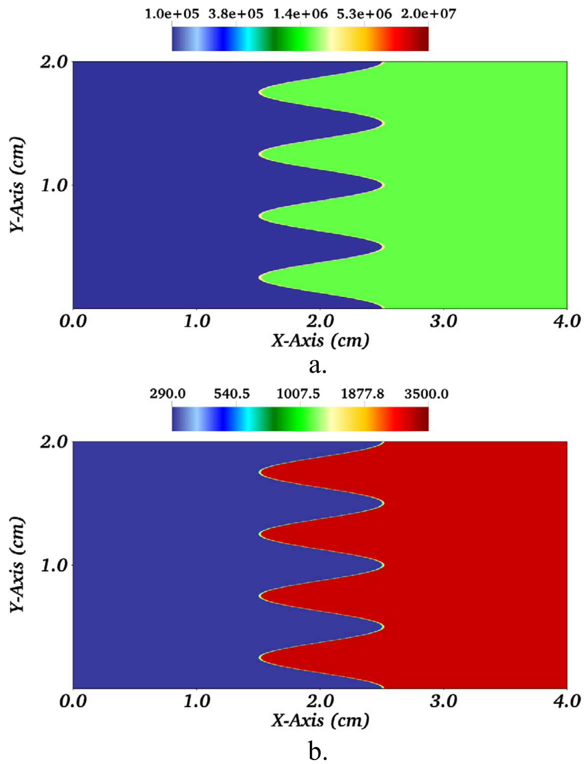


Fig. 8. The initial conditions for a two-dimensional detonation, showing a flow field based on a one-dimensional ZND profile perturbed with a sine function in the direction of propagation. (a) Pressure, Pa. (b) Temperature, K.

$$\frac{dY}{dt} \equiv \dot{w} = -A\rho Y \exp\left(-\frac{Q}{RT}\right), \tag{44}$$

where A is the pre-exponential factor, Q is the activation energy, and R is the universal gas constant. The numerical methods used to solve the equations are the same as those described above, except that piecewise linear method (PLM) [42] is utilized for state reconstruction, instead of PPM that tends to exhibit carbuncle phenomenon for strong shocks such as the ones present in a detonation. Chemical source terms are coupled to the flow equations via Strang splitting [83].

The different parameters and flow properties for the reaction model are summarized in Table 2. This model is based on the single-step reaction mechanism developed by Gamezo et al. [84]. Originally, the model parameters were calibrated to reproduce both one-dimensional laminar flame and Zel'dovich, von Neumann, Döring (ZND) detonation solutions for a stoichiometric hydrogen-air mixture. Taylor et al. [76], however, have shown that this model cannot successfully predict the cellular structure of a detonation wave. In particular, it leads to an incorrect temperature profile and an extremely irregular cell structure. They suggested to modify the model by lowering the values of both the activation energy, Q, and the pre-exponential factor, A, in order to obtain a more regular cellular structure. The chosen values maintain the correct half-reaction thickness, yet, they cannot reproduce the correct laminar flame solution. As the current problem of interest involves only detonations, this deficiency should not affect the results. It is stressed that the changes of the original

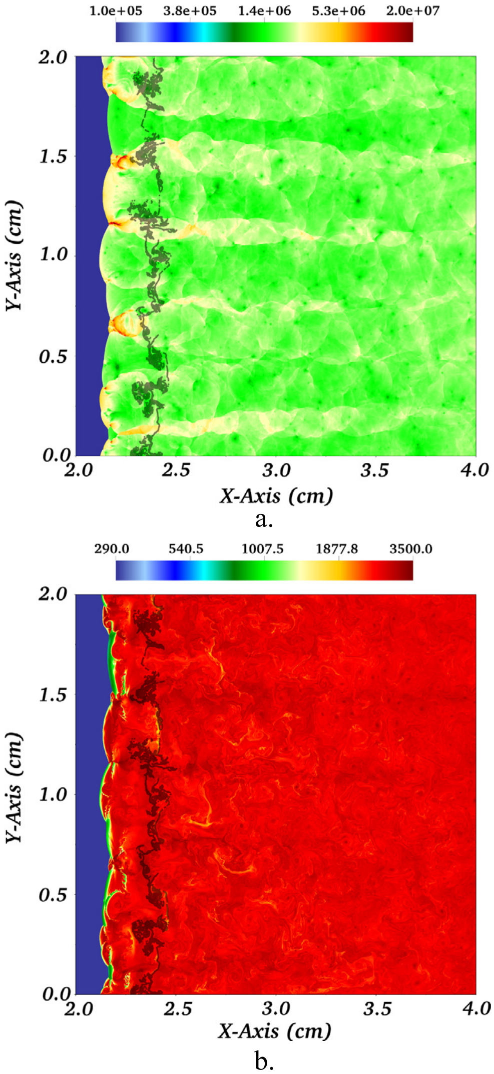


Fig. 9. Fully developed two-dimensional cellular detonation. The flow field is shown after 3.3 flow-through times. Final particle positions are marked as black dots: (a) Pressure, Pa. (b) Temperature, K.

model have no effect on the Chapman-Jouguet (CJ) detonation velocity or the post-shock conditions. Accordingly, one of the suggested modified parameter sets was implemented in our simulations, as shown in Table 2.

The problem is solved in a 4 × 2 cm rectangular domain with 32768 × 16384 equally spaced grid cells that comprise in total about half a billion cells. The boundary conditions are no-slip and no-penetration for the upper and lower boundaries and zero-order extrapolation (outflow) for the left and right boundaries. The initial flow field is based on the 1D ZND profile and the entire problem is solved in the shock frame of reference. The post-shock values that represent the first data point in the ZND profile are located at the center of the domain and they have an x-direction displacement according to a sine function (see Fig. 8a and Fig. 8b for the initial pressure (Pa) and temperature (K) contours, respectively). Gas ahead of the shock has the initial thermodynamic property values as shown in Table 2, and it moves with the CJ detonation velocity to the right. All the property values after the end of the reaction zone are constant according to the last value of the ZND profile. As the initial front is curved, it becomes unstable and a multi-cellular detonation arises. It should be noted that the initial ZND profile is calculated according to the original model of Gamezo et al. [84] and not the modified one. As mentioned above, this does not change the post-shock state, but it does change the initial profile. This initial structure is only used to initiate a detonation, which after the initial transient develops a fully multi-dimensional structure. After the initial conditions are set, the flow equations are integrated for about 3.1 flow-through times, where one flow-through time is defined as the domain length, L, divided by the CJ detonation velocity, $\tau_d = L/D_{CJ}$. This time duration allows for the multidimensional detonation structure to become fully developed. Then, a total of 49,152 tracer point-particles are injected at $x = 2$ cm. The particles initially comprise three sequential columns, spanning the entire height of the domain, with each particle seeded at a cell center. Finally, the entire set of flow and particle equations is integrated for 0.2 flow through time.

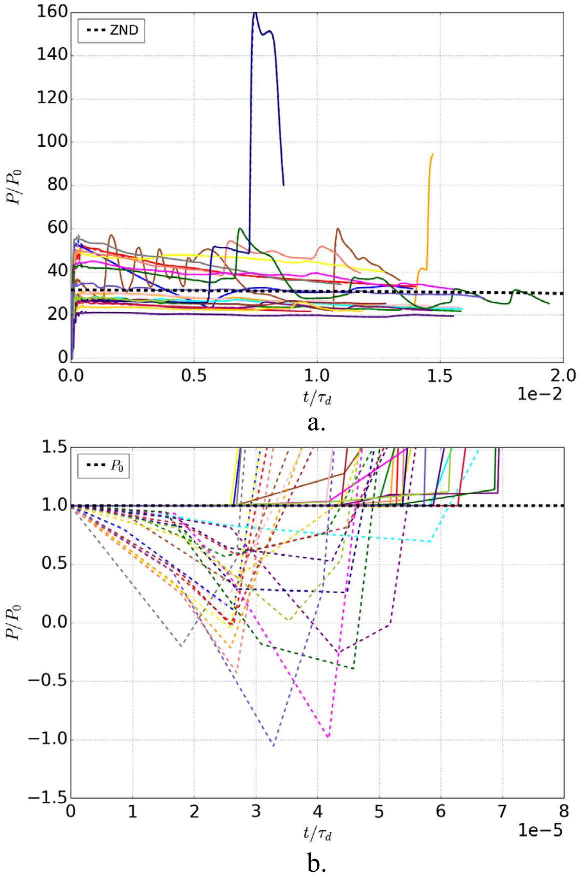


Fig. 10. Twenty sample particle trajectories. Normalized interpolated pressure is shown against the normalized time. Results for WENO-5 (solid lines) and W_5 (dashed lines) interpolation schemes are presented: (a) The entire time duration. (b) A magnified view of the time when particles are crossing the shock.

Figs. 9a and 9b show pressure and temperature field in the domain, respectively, at the end of the simulation with the particle final positions marked as black dots. One can see that the detonation front is corrugated as transverse waves propagate in the y-direction and interact with the leading shock. This creates triple shock points, where the gas is highly compressed, and which divide the front into cells. The results qualitatively agree with those in [76], where the same one-step reaction kinetics mechanism was utilized. Note that the current simulation resolution is much higher, and the grid cell size is 8 times lower than the one used in [76]. We tested the problem with lower resolutions, with the fully developed detonation structure found to be fully converged. The transient process, however, was found to be very different between the high- and low-resolution cases. In particular, at the highest resolution, initially the leading shock decoupled from the reaction front followed by the deflagration-to-detonation transition (DDT). Although this transient process is not relevant for the current study, it deserves further analysis in the future work.

Lagrangian-particle analysis described above is carried out using the high-order interpolation methods, W_5 and WENO-5, in order to compare their performance. As mentioned above, high-order interpolation is preferred due to its superior accuracy. The entire analysis is performed for a data set that includes the particle interpolated pressure and temperature approximately between 3.13 and 3.16 flow-through times since the start of the simulation. This time interval starts slightly before the first column of particles enters the leading shock and ends slightly after the last of column of particles leaves the reaction zone. Fig. 10a shows a sample of 20 particle trajectories, for W_5 (dashed lines) and WENO-5 (solid lines), respectively, where interpolated pressure normalized by the initial pressure, P_0, is shown against time normalized by the flow-through time, τ_d. The time series are all shifted, according to the results of W_5, such that at $t = 0$ s, particles first reach a pressure different than the initial one. Also, the ZND pressure profile is shown with a dashed black line.

For both interpolation schemes, the trajectories are quite irregular and can be very different. For instance, pressure spikes to values over 150 bar for one of the trajectories, which is about 3 times higher than the maximum pressure for most of the presented trajectories, and about 5 times higher than the ZND pressure. Comparison between the trajectories reproduced by the two interpolation schemes shows that the trajectories differ in one important respect. In particular, Fig. 10b shows a magnified view of the time the particle enters the shock region. One can see that in the case of W_5, for all the trajectories,

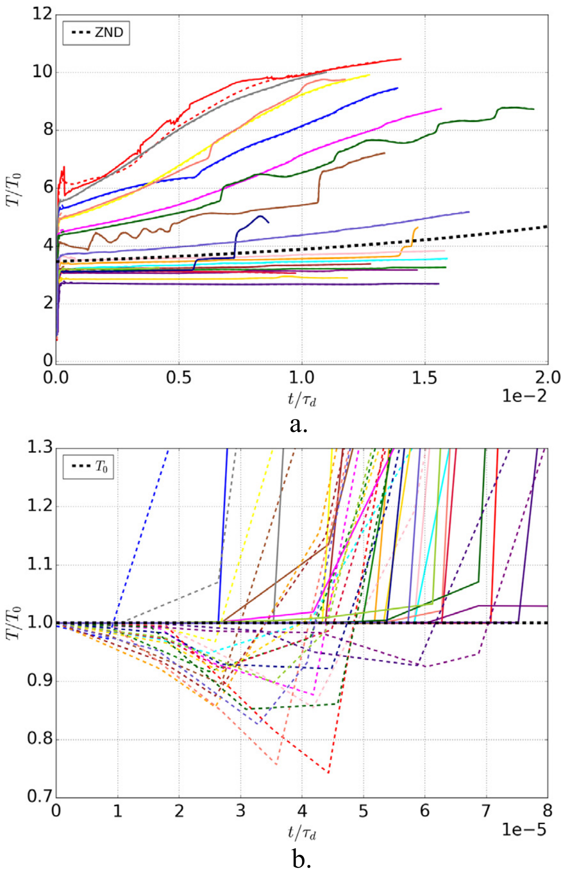


Fig. 11. Twenty sample particle trajectories. Normalized interpolated temperature is shown against the normalized time. Results for WENO-5 (solid lines) and W_5 (dashed lines) interpolation schemes are presented: (a) The entire time duration. (b) A magnified view of the time when particles are crossing the shock.

the normalized pressure values fall below its initial value of unity, which is unphysical. For some of the trajectories, the pressure becomes negative, with a value as low as −1. On the other hand, for WENO-5, the magnified view shows that for all sampled trajectories, the normalized interpolated pressure is always higher than its initial value, and no numerical oscillations are observed. Fig. 11a is similar to Fig. 10a showing temperature normalized by the initial temperature, T_0. The normalized interpolated temperature trajectories are also highly irregular and differ from the ZND temperature with a maximum value higher by almost 3 times. Fig. 11b shows a magnified view of the time the particle enters the shock. Again, the results for W_5 are lower than the initial normalized interpolated temperature of unity, with values almost as low as 0.7, which is unphysical. For WENO-5, all the normalized interpolated temperatures are higher than unity and no visible oscillations are observed.

Fig. 12 shows normalized interpolated pressure vs. temperature for WENO-5 and the same 20 particle trajectories presented in the figures above. The dashed black line corresponds to the ZND profile. This graph illustrates the way a joint probability density function (PDF) shown below is constructed from different particle trajectories. Each data point represents a time that a Lagrangian particle attained a certain value of pressure and temperature. By using all data points from the numerical simulation results, a joint PDF can be constructed. Figs. 13a and 14a show joint PDFs for W_5 and WENO-5, respectively, where the color bar denotes the natural logarithm of the joint PDF, and the black dashed line corresponds to the ZND profile. The two PDFs show very similar trends, although they are not identical. For both PDFs, normalized pressure and temperature can change significantly along the particle trajectory and reach values higher than 200 and 11, respectively, which are very different from the ZND profile. The main difference between the two PDFs is in the low pressure and temperature region corresponding to the initial shock crossing. For W_5 (see Fig. 13a), this region is much broader in comparison to the same region in the WENO-5 PDF (see Fig. 14a). Results again reveal non-physical trends once W_5 is utilized for interpolation. A magnified view of the low pressure and temperature region for W_5 is shown in Fig. 13b. One can see that the normalized pressure becomes negative with values as low as about −2. Normalized temperature also drops below its initial value to values as low as about 0.7. On the other hand, for WENO-5 (see Fig. 14b) the magnified view shows that for all the trajectories, interpolated pressure and temperature values are always higher than their initial value. This result shows that for the Lagrangian-particle tracking analysis of highly compressible reactive flows, high-order centered schemes,

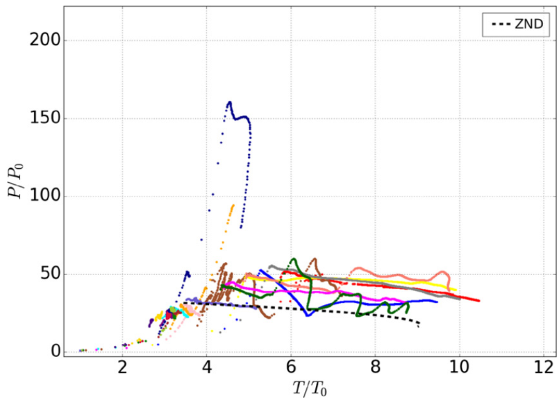


Fig. 12. Twenty sample particle trajectories. Normalized interpolated pressure is shown against the normalized temperature for the WENO-5 interpolation scheme.

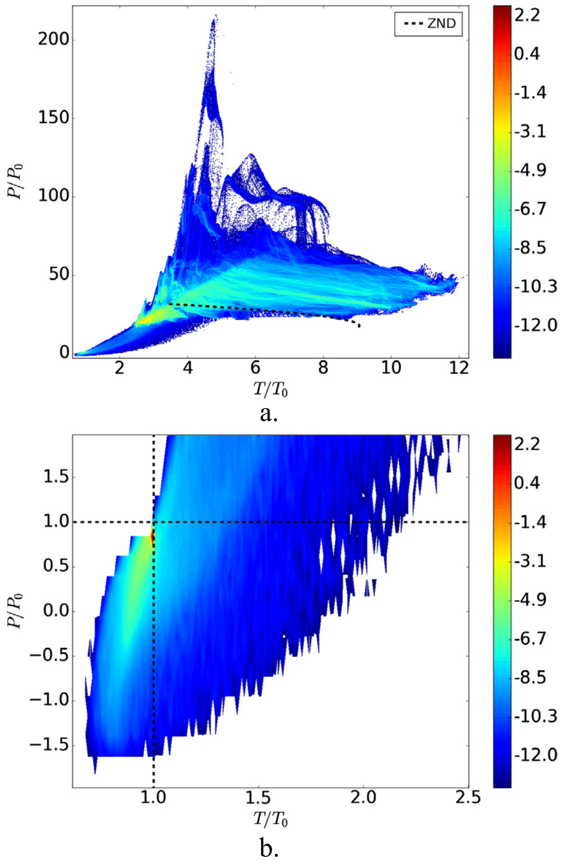


Fig. 13. Joint PDF of the normalized interpolated pressure and temperature for all particle trajectories. The color bar denotes the natural logarithm of the joint PDF. Values are reconstructed with the W_5 interpolation scheme: (a) The full data set. (b) A magnified view for the low pressure and temperature region.

like W_5, can lead to unphysical thermochemical trajectories, whereas WENO schemes provide an alternative with the same order of accuracy, yet free of severe numerical oscillations and unphysical numerical artifacts.

4. Conclusions

In the current work, we present WENO-3 and WENO-5 interpolation schemes for the purpose of Lagrangian-particle advection in highly compressible flows. Extensive set of tests is described, including comparison against various traditional centered interpolation schemes. The interpolation accuracy is first tested for simple two-dimensional vortex flows. Two

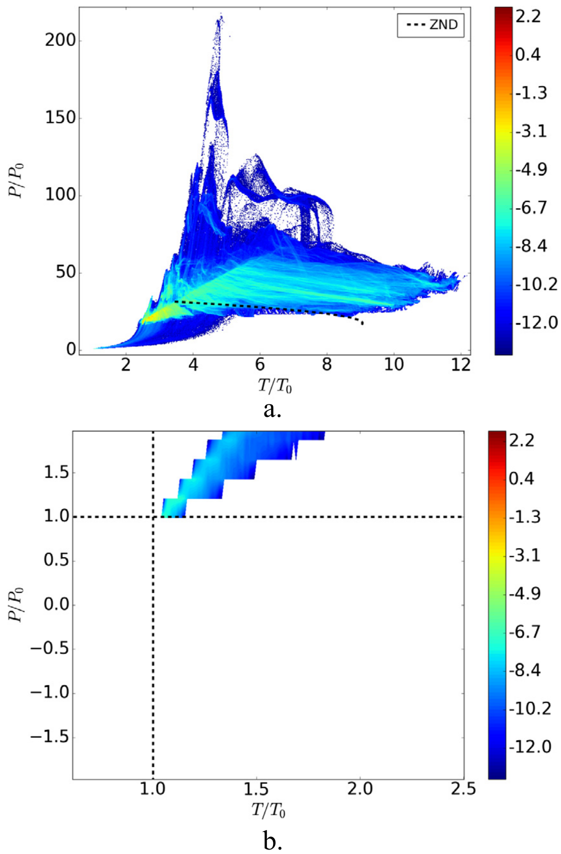


Fig. 14. Joint PDF of the normalized interpolated pressure and temperature for all particle trajectories. The color bar denotes the natural logarithm of the joint PDF. Values are reconstructed with the WENO-5 interpolation scheme: (a) The full data set. (b) A magnified view for the low pressure and temperature region.

cases have been investigated, namely a uniform vortex and a shocked vortex with a velocity discontinuity. For smooth flows, it is demonstrated that WENO schemes provide accuracy comparable to high-order centered schemes with the same order of accuracy. At the same time, once discontinues are introduced into the flow, it is shown that the maximum relative interpolation errors of centered schemes in the vicinity of the discontinuity can be larger than 100%. On the other hand, the maximum relative interpolation errors for WENO-3 and WENO-5 are about 1% and 0.4%, respectively.

The benefits of WENO interpolation over centered schemes were further explored by interpolating pressure, temperature, and density at the particle position for a Mach 10 normal shock wave. It is shown that low-order schemes, like CIC, smear the shock structure. High-order interpolation schemes, like W_5, not only smear the shock, but also produce numerical oscillations that can lead to unphysical results, such as negative absolute pressure values. It is explained that the origin of the oscillations lies in the negative weights that are inherent for any weight function, without special treatment, with an order of accuracy greater than 2nd. For WENO schemes, the shock is captured in a much sharper manner than for centered schemes, and no visible oscillations occur. A similar test is then performed for an unsteady three-dimensional spherical blast wave, where the shock is reconstructed by a shock capturing method, and thus it is broadened over several grid cells. The results show that qualitatively the same trends as for the normal shock test are observed. Thus, WENO schemes are advantageous even for more complex compressible flow fields.

The final test demonstrates Lagrangian-particle tracking analysis of a highly compressible reactive flow, namely, two-dimensional cellular gas-phase detonation. The two high-order interpolation methods, W_5 and WENO-5, are used to reconstruct pressure and temperature at the particle position. Comparison between the two includes several sampled particle trajectories and joint PDFs that are constructed according to each interpolation scheme. The results show that for all sampled trajectories, W_5 leads to interpolated pressure and temperature values that are lower than their initial value, which is physically impossible. For WENO-5, the interpolated values are always equal to or higher than their initial value. Comparison between the two joint PDFs reveals that the low normalized pressure and temperature regions are significantly different. It is shown that for W_5, normalized pressure and temperature values can drop as low as -2 and 0.7, respectively, which is not possible physically. For WENO-5, all the 49,152 particle trajectories do not exhibit any non-physical trends. It is concluded that for the Lagrangian-particle tracking analysis of highly compressible reactive flow regimes, high-order centered interpolation schemes, like W_5, can lead to numerical oscillations and non-physical results that may produce incorrect PDFs.

WENO schemes, like WENO-5, are shown to provide high-order interpolation without any severe numerical oscillations and non-physical artifacts.

WENO interpolation for Lagrangian particle advection is a robust numerical tool for highly compressible flows, where low-order centered interpolation schemes cannot provide a sufficient level of accuracy and high-order centered interpolation schemes tend to oscillate. Although the current work is focused on massless point particles, this interpolation method can be used for particles with mass as well. These might involve more complex physics like two-way coupling with the gas, as well as chemically reacting particles. In general, WENO interpolation can be found advantageous for a variety of practical applications, including high-fidelity direct numerical simulations of multi-physics flows in supersonic and hypersonic aerodynamics, and in advanced airbreathing hypersonic and detonation-based propulsion systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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